SYSTEM BRIEF

JobTrace

A job-application tracker that runs entirely on your own computer — what it is, how to run it on a Mac or a PC, everything it does, and how it works underneath.

CONTENTS

Runs on macOS & Windows **No account · no cloud · no cost** **For** anyone using or evaluating JobTrace

01 What JobTrace is

A job search generates a mess of state: dozens of applications, each at a different stage, each with its own thread of emails, assessments, and interviews. JobTrace is a private dashboard that holds all of it in one place.

The important word is **private**. There is no website to sign into, no account, no subscription, and no company on the other end. JobTrace is a small program that lives in a folder on your computer. When you open it, it shows a dashboard in your web browser — but that dashboard is served from your own machine, not from the internet. Everything you type stays in a single file on your disk.

> **The whole picture.** Your browser and the JobTrace program are two things on your computer talking to each other; the program keeps everything in one file.

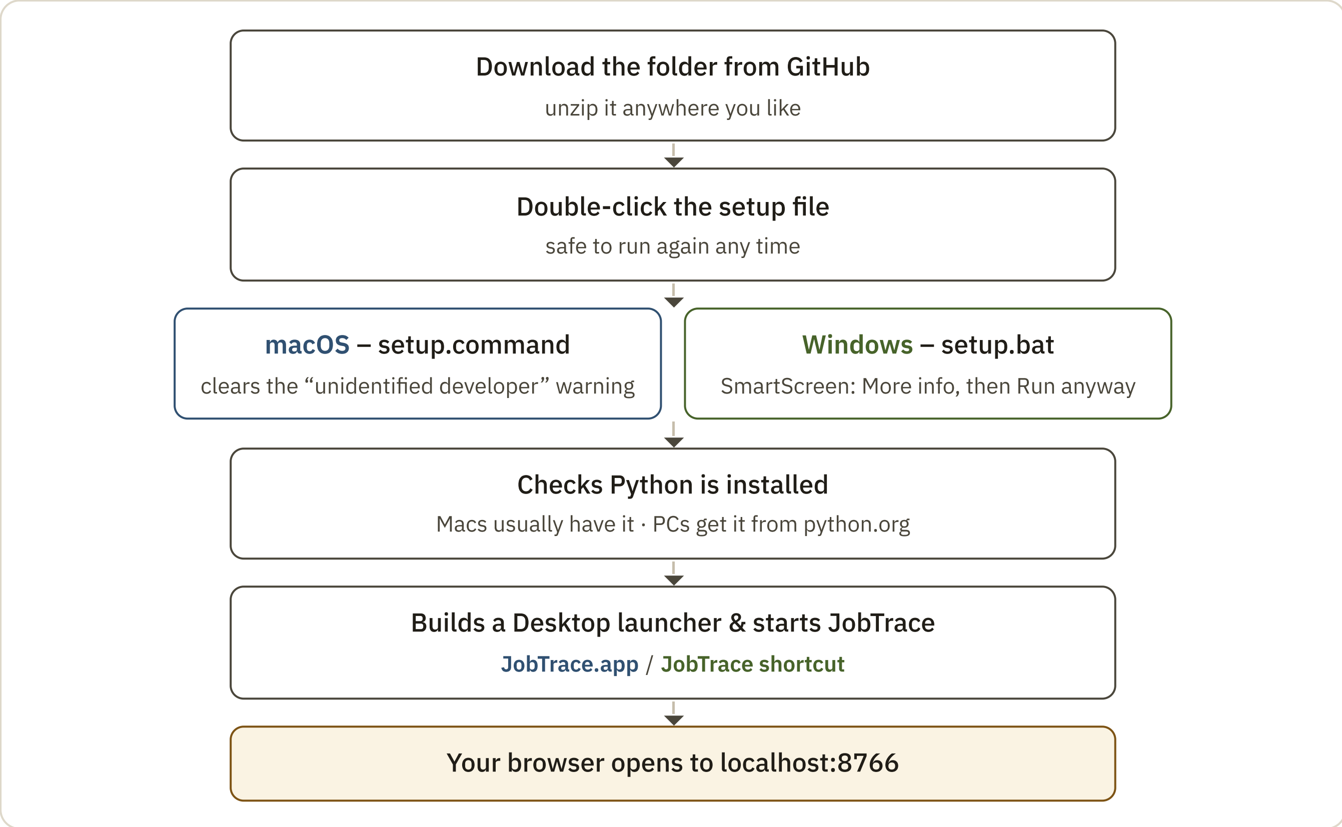
Because it is just a folder and a file, JobTrace is easy to reason about: to back it up, copy the file; to move it to another computer, copy the folder; to delete it forever, drag the folder to the trash. Nothing is left on a server somewhere.

02 Installing & running it

THE EASIEST WAY

Open the folder with an AI coding assistant — Claude, Codex, or similar — and ask it to *"set up JobTrace."* It reads the setup guide, checks what your computer needs, starts the program, and can walk you through connecting Gmail if you want it. You do nothing but answer the occasional question.

Prefer to do it by hand? It is one download and one double-click. The double-click runs a small script that does the fiddly parts — checking that a piece of free software called **Python** is present, creating a launcher on your Desktop, and starting the program.



> **First run, start to finish.** The path is the same on both systems; the two coloured boxes are the only places `macOS` and `Windows` differ. `localhost:8766` is an address that exists only on your own machine.

1. **Get the folder.** Download it from GitHub and unzip it. Put it wherever you keep your files — just *not* inside a cloud-synced folder (see the note below).
2. **Double-click the setup file.** `setup.command` on a Mac, `setup.bat` on Windows. Your system may ask you to confirm you trust it — normal for a downloaded file.
3. **If it asks for Python,** install it from python.org and run the setup file again. On Windows, tick **“Add python.exe to PATH”** during that install.
4. **You’re done.** A JobTrace launcher appears on your Desktop and the dashboard opens in your browser. Drag the launcher to your Dock or taskbar.

Every day after that

Double-click the JobTrace launcher. Nothing flashes on screen; after a second or two your browser opens to the dashboard, and the program keeps running quietly in the background. To stop it, double-click `stop.command` (Mac) or `stop.bat` (Windows). To start it automatically at login, add the launcher to your login items (Mac) or Startup folder (Windows).

KEEP IT OFF CLOUD DRIVES

If the JobTrace folder sits inside iCloud Drive, OneDrive, Google Drive, or Dropbox, your data file gets uploaded to that service like any other file — which quietly breaks the “stays on your computer” promise. Keep the folder somewhere ordinary, like your home folder.

03 What it does

JobTrace has three screens — **Dashboard**, **Tracker**, and **Analytics** — plus the tools to get data in and out.

Dashboard
Every application in one searchable, sortable, filterable table. Above it, summary tiles: how many you've sent, how many replied (split positive / negative), how many reached an interview, how many became offers — plus your response and interview conversion rates, and a few charts.

Tracker
The two things email can't tell you on its own. A **to-do list of assessments** you've been invited to but haven't finished, each with an optional deadline and a "mark completed" button. And your **interviews** as a list or a month calendar — you set the exact date and time, and you record the outcome yourself (offer, next round, or not selected), because recruiters often never put that in writing.

Analytics
Applications over time, a response funnel from "applied" down to "offer," how your applications spread across stages and outcomes, and a table showing which sources actually convert for you.

Each application in detail
Company, position, location, source, dates, the job description, your notes, the job link — plus a full **timeline** of what happened and when, with entries added automatically whenever you move an application forward.

Import, export, back up
Bring applications in from a spreadsheet (checked all-or-nothing, so you never get a half-imported mess). Export everything to CSV or JSON. Make a safe, timestamped copy of your database with one click.

How an application moves

JobTrace keeps two things separate: the **stage** an application is at, and its **outcome**. That way "I got an assessment invite" and "I got rejected" are both recordable without cramming them into one status.

`Applied` → `Screening` → `Assessment` → `Interview 1 / 2` → `Final interview` → `Offer`

`Closed — from any point`

at `Assessment`	at `Interview`
the Tracker's to-do list picks this up	you set the date & record the result

> **An application's life.** It moves left to right through the stages. The **Tracker** lifts out the two stages that need your hands-on attention. **Closed** is an exit you can take at any point.

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04 The optional Gmail sync

Left to itself, JobTrace never touches the internet. If you want, you can let it **read** your inbox for recruiter replies and update the tracker for you — new confirmations become applications, rejections close them out, interview invites move them forward. It only ever reads; it never sends, deletes, or files anything.

Three ways to run a sync, chosen in the dashboard under the gear icon next to the Gmail label:

A **Ask your assistant — easiest, nothing to set up**

Open the folder with an AI coding assistant and say "sync my Gmail." The assistant is the connection; it reads your inbox and updates JobTrace right then.

`Setup` none `Runs` when you ask `Cost` your assistant plan

B **Scheduled, via a command-line tool**

Install the assistant's CLI, sign in, and connect Gmail *to that tool*. JobTrace then runs it on a timer. The dashboard shows a checklist of anything still missing.

`Setup` sign in + connect Gmail `Runs` on a timer `Cost` your assistant plan

C **Scheduled, with your own keys**

Paste in a Gmail app password (or a Google sign-in) and your own Anthropic API key. No assistant app involved.

`Setup` two pasted secrets `Runs` on a timer `Cost` a few cents per run, billed to you

A · ask your assistant

B · scheduled via a CLI tool

C · scheduled with your keys

→ all write

The same files in `data/`

the dashboard updates the same way

> **Three routes, one destination.** Whichever you choose, the result lands in the same place — so you can start with **A** and move to **C** later without anything breaking.

ONE THING TO CHECK

A scheduled sync reads whichever mailbox your assistant is signed into. The dashboard shows the connected address on the Gmail status screen — glance at it once to be sure it's the inbox you meant.

Automatic sync only runs while JobTrace is open. The timer lives inside the program, so keep the launcher in your login items if you want it to stay current in the background.

05 macOS vs Windows, side by side

JobTrace behaves the same on both systems. The differences are all in the plumbing of getting it started, and they are small.

	MACOS	WINDOWS
Setup file	`setup.command`	`setup.bat`
Python	Built in on most Macs (3.9 or newer). Only visit python.org if setup says it's missing.	Usually not installed. Get it from python.org and tick **"Add python.exe to PATH"** during the install.

	MACOS	WINDOWS
First-open prompt	"Unidentified developer." The setup file clears this; if it still blocks, right-click → **Open** → **Open** once.	"Windows protected your PC" (SmartScreen). Click **More info** → **Run anyway**.
Launcher created	`JobTrace.app` on the Desktop. Drag it to the Dock.	A `JobTrace` shortcut on the Desktop. Pin it to the taskbar.
While running	No window at all — it runs quietly in the background on both.	
Start / stop by hand	`start.command` / `stop.command`	`start.bat` / `stop.bat`
Run at login	System Settings → General → Login Items	Put the shortcut in the Startup folder (`shell:startup`)
Moving the folder	The launcher remembers where the folder was — re-run the setup file after moving it.	

THE REST IS OPTIONAL — SKIP IT UNLESS YOU'RE CURIOUS HOW IT WORKS

06 Under the hood

None of this matters for using JobTrace. It is here for anyone who wants to understand what they are running, or who might want to change it.

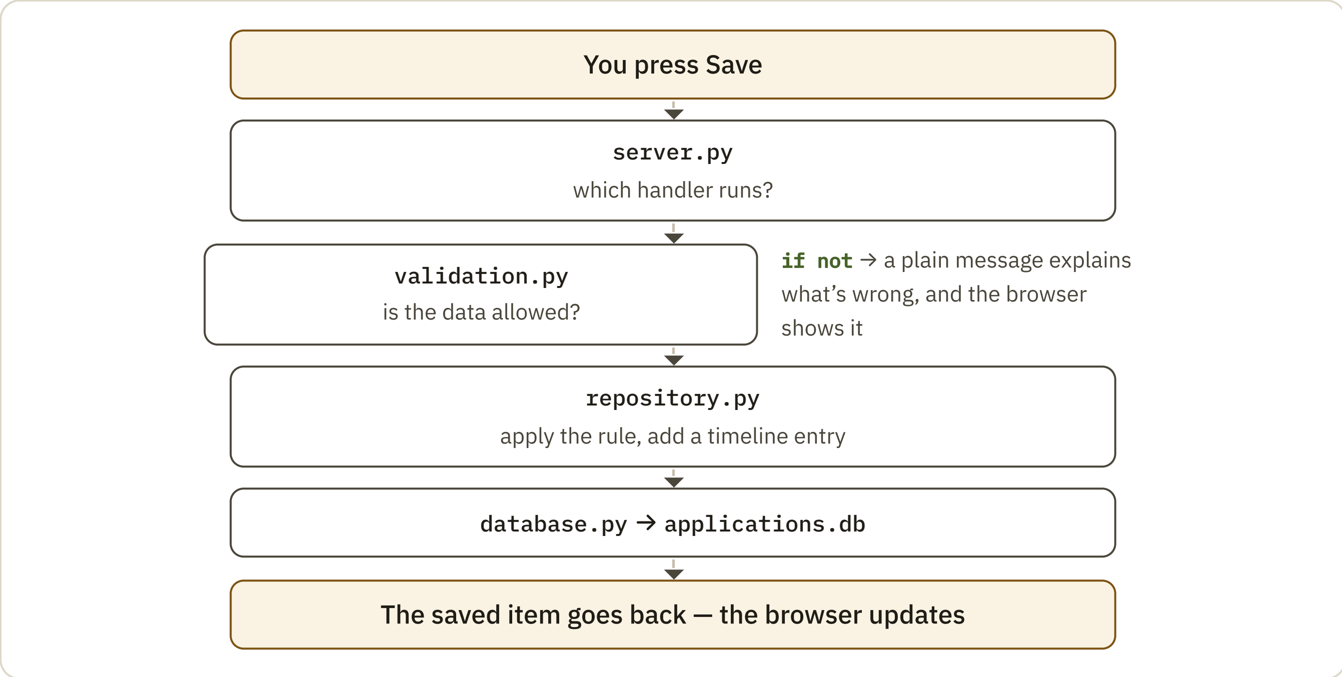
One small program and one file

JobTrace is written in **Python**, using only the language's built-in toolkit — no frameworks, no packages to install, nothing to compile. When it starts, it opens your data file and begins listening on the address `localhost:8766`, waiting for your browser to ask it for something.

The dashboard is plain HTML, CSS, and JavaScript, served straight from the folder — no build step. All of your data lives in `data/applications.db`, a **SQLite** database: a whole database that is just one file, the same format used inside phones and browsers.

What happens when you press "Save"

Every change follows the same short path, and each step is a small file with one job. The browser never writes to the database directly — it always goes through the rules.



> **The request lifecycle.** `server.py` only sorts the mail. `repository.py` is where every rule lives — what a stage change means, how the stats are counted, when a timeline entry is added. Bad input is turned away at `validation.py` with a plain explanation, never a crash.

The folder, briefly

`backend/`	The program: `server.py` → `repository.py` → `validation.py` → `database.py`, in order of "closeness to the data."
`backend/sync/`	Gmail sync: the timer that runs it, the piece that reads Gmail, and the settings.
`frontend/`	The dashboard — one HTML file, one stylesheet, and a handful of small JavaScript files, one per screen.
`data/`	`applications.db` (everything), `sync_config.json` (sync settings), a couple of small bookkeeping files, and sync logs. Never shared, never in version control.
`backups/`	Timestamped copies of the database — made when you click "back up," and automatically before any change to the database's structure.
`tests/`	Automated checks — `python3 -m unittest discover tests` — that confirm the rules, the maths, and every screen still work after a change.

Why it's built this way

The design goal was **durability**, not features. Standard-library Python means it will still run in ten years with no dependency to have rotted. One SQLite file means backup and recovery are "copy a file." All the rules in one place (`repository.py`) means the dashboard and any future automation can never disagree about what the numbers mean. And an additive-only approach to the database means upgrading never rewrites your existing data — new columns are added alongside the old, with a backup taken first.

JobTrace runs on your computer. There is no cloud service, no account, and nothing leaves your machine unless you deliberately connect Gmail — which is read-only. This brief describes the current version; the project's `README.md` and `SETUP.md` carry the exhaustive detail.